What happens to a coefficient when we add a control?

Empirical Methods for Law and Economics | EDLE

Supplementary Material

Marco Rosso
University of Bologna

The question

- We estimate the association between an outcome Y and a variable of interest X.
- We then add a control Z. The coefficient on X changes.
- **Question from class:** can it go *up* as well as *down*?

Short answer: yes. It can fall, rise, change sign, or stay exactly the same.

Which one happens is not a matter of luck: it follows from an **exact algebraic identity** that we can compute in any dataset.

Two regressions and one auxiliary regression

$$\begin{aligned} \text{Short:} \quad & Y_i = a + \beta^s X_i + e_i \\ \text{Long:} \quad & Y_i = a' + \beta^l X_i + \gamma Z_i + e_i' \\ \text{Auxiliary:} \quad & Z_i = c + \delta X_i + w_i \end{aligned}$$

- γ: how Z relates to Y, holding X fixed.
- δ: how Z moves with X (the slope of Z on X).
- All three are estimated by OLS on the *same* observations.

The omitted variable identity

$$\boxed{\hat{\beta}^{s} \;=\; \hat{\beta}^{l} \;+\; \hat{\gamma}\,\hat{\delta}} \qquad \Longleftrightarrow \qquad \hat{\beta}^{l} - \hat{\beta}^{s} \;=\; -\,\hat{\gamma}\,\hat{\delta}$$

- In OLS this is an **identity**, not an approximation: it holds to the last decimal in every sample.
- Adding Z moves the coefficient on X by exactly $-\hat{\gamma}\hat{\delta}$.
- If the long regression also contains other controls W, the auxiliary regression must contain them too: Z on X *and* W.
- With several new controls $Z_1, \ldots, Z_K$ the shift is $-\sum_k \hat{\gamma}_k \hat{\delta}_k$.

Reading the direction from two signs

sign of $\hat{\gamma}$	sign of $\hat{\delta}$	adding $Z\ldots$
$+$	$+$	lowers the coefficient on X
$-$	$-$	lowers the coefficient on X
$+$	$-$	**raises** the coefficient on X
$-$	$+$	**raises** the coefficient on X
0 or	0	leaves it unchanged

- If $|\hat{\gamma}\hat{\delta}| > |\hat{\beta}^{\prime}|$ and the shift goes against $\hat{\beta}^{\prime}$, the coefficient **changes sign**.
- A control that raises the coefficient is often called a *suppressor*: leaving it out was hiding part of the association.

PREDICT Three simulated worlds

In all three worlds the true model is the same:

$$Y = 1 \cdot X + 2 \cdot Z + u, \qquad X,\ v,\ u \sim N(0,1) \text{ independent}$$

Only the link between Z and X changes:

World	How Z is generated	Short coefficient $\hat{\beta}^s$?
A	$Z = \ \ 0.5\,X + v$	?
B	$Z = -0.5\,X + v$	?
C	$Z = -1\,X + v$	?

Before running anything: what does the regression of Y on X alone give in each world, and in which direction does it move once we add Z?

ANSWER Three simulated worlds

The long regression recovers $\beta^l \approx 1$ and $\gamma \approx 2$ in all three worlds. The short one gives $1 + 2\delta$:

World	δ	$\gamma\delta$	$\hat{\beta}^s \approx$	Adding Z...
A	0.5	1	2	halves the coefficient (down)
B	-0.5	-1	0	reveals an effect that was invisible (up)
C	-1	-2	-1	flips the sign (from negative to positive)

- Same true effect of X, three very different short regressions.
- World B is the most dangerous in practice: a zero in the short regression looks like "no effect".

STATA Check the identity yourself (simulated data, world A)

```
clear all
set seed 2026
set obs 10000
generate x = rnormal()
generate v = rnormal()
generate u = rnormal()
generate z = 0.5*x + v              // world A: change 0.5 to -0.5 or -1
generate y = 1*x + 2*z + u

quietly regress y x                 // short regression
scalar b_short = _b[x]
quietly regress y x z               // long regression
scalar b_long  = _b[x]
scalar gamma   = _b[z]
quietly regress z x                 // auxiliary regression
scalar delta   = _b[x]

display "short - long = " b_short - b_long "   gamma*delta = " gamma*delta
```

The two numbers printed on the last line coincide exactly. Rerun with worlds B and C.

Real example 1: returns to schooling and ability (down)

- Y = log wage, X = years of schooling, Z = a measure of ability (e.g. the AFQT test score in the NLSY).
- $\gamma > 0$: higher ability, higher wages at a given schooling level.
- $\delta > 0$: more able people stay longer in school.
- Adding the ability measure **lowers** the estimated return to schooling: the short regression partly credits schooling for ability ("ability bias").
- The debate on how large this bias is runs from Griliches (1977) to Card (1999).

Griliches, Z. (1977), Econometrica. Card, D. (1999), Handbook of Labor Economics, Vol. 3A.

Real example 2: the gender wage gap (up and down)

- Y = log wage, $X = 1$ if woman. The coefficient is negative (the raw gap).
- **Controlling for education raises the unexplained gap.** In recent US data women are on average *more* educated than men ($\delta > 0$) and education raises wages ($\gamma > 0$). The shift is $-\gamma\delta < 0$: the coefficient becomes more negative.
- **Controlling for actual work experience shrinks it.** Women have on average fewer years of experience ($\delta < 0$) and experience raises wages ($\gamma > 0$): the shift is positive, toward zero.
- Same outcome, same X: the direction depends on the control.

Blau, F. D. and L. M. Kahn (2017), "The Gender Wage Gap: Extent, Trends, and Explanations", Journal of Economic Literature.

Real example 3: Berkeley admissions, 1973 (sign flip)

- Aggregate data on graduate admissions: women were admitted at a visibly lower rate than men.
- Conditioning on the **department** of application, the gap disappears; in several departments it reverses slightly in favour of women.
- Mechanism: women applied disproportionately to departments with low admission rates for everybody.
- In our notation: department selectivity affects admission ($\gamma < 0$) and is correlated with being a woman ($\delta > 0$). The shift $-\gamma\delta$ is positive and large enough to change the sign.
- This is **Simpson's paradox**: the extreme case of the identity.

Bickel, P. J., E. A. Hammel and J. W. O'Connell (1975), "Sex Bias in Graduate Admissions: Data from Berkeley", Science.

Real example 4: the Black–white wage gap (large drop)

- Y = log wage, $X = 1$ if Black (NLSY, young men).
- Z = AFQT score measured *before* entry into the labour market.
- Adding Z **reduces the gap substantially** for men.
- Why the timing of Z matters: a score measured as a teenager is not itself a consequence of labour market discrimination later on. The authors are explicit that it may still reflect earlier disadvantages.

Neal, D. A. and W. R. Johnson (1996), "The Role of Premarket Factors in Black-White Wage Differences", Journal of Political Economy.

DISCUSS Police and crime across municipalities

A researcher regresses the crime rate of each municipality on the number of police officers per capita. A colleague suggests adding average **income**.

- Write down the sign you expect for γ (income $\rightarrow$ crime, holding police fixed).
- Write down the sign you expect for δ (police on income).
- In which direction should the police coefficient move?

Is there more than one defensible answer? What would you need to know about these municipalities?

ANSWER Police and crime across municipalities

- There is no single correct sign: the point of the exercise is that the answer depends on facts about the sample.
- **One story:** richer places have less crime ($\gamma < 0$) and fund more police ($\delta > 0$). Then $-\gamma\delta > 0$: adding income moves the police coefficient up, toward zero or above it.
- **Another story:** rich places offer more to steal (property crime, $\gamma > 0$). Then the shift goes the other way.
- The identity tells us *what to check*: estimate δ (one auxiliary regression) and look at γ in the long regression.
- Note that income does not solve the main problem: police are assigned where crime is high, so the coefficient is not causal with or without the control.

DISCUSS Should we control for occupation?

We want the effect of a university degree on wages. A colleague says: "Add occupation dummies; it makes the estimate more precise and more credible."

Do you agree? What happens to the degree coefficient, and how should we interpret it?

ANSWER Should we control for occupation?

- Occupation is partly an **outcome** of the degree: a degree is one way people get into better-paid occupations.
- Controlling for it removes that channel: we compare graduates and non-graduates *within* the same occupation, which is a different question.
- Worse, the comparison is not clean: non-graduates who become managers are probably different (in ability, contacts...) from graduates who become managers. Conditioning on occupation creates a new selection bias.
- This is a **bad control** (Angrist and Pischke, 2009, Section 3.2.3). A good control is determined before X and is not affected by it.
- The identity still holds; what changes is that the long coefficient is no longer the one we want.

Angrist, J. D. and J.-S. Pischke (2009), Mostly Harmless Econometrics, Princeton University Press.

Three warnings for applied work

1. **Stability is not proof of no bias.** With several controls the shifts $\gamma_k \delta_k$ can offset each other: an unchanged coefficient may hide two biases in opposite directions.
2. **Order dependence.** If controls are added one at a time, how much "movement" we attribute to each depends on the order. Gelbach (2016) gives an order-invariant decomposition (`b1x2` in Stata).
3. **Scale the movement.** A coefficient that barely moves when the controls explain very little of Y is weak reassurance. Oster (2019) formalises this by combining coefficient movement with the change in R^2 (`psacalc` in Stata).

Gelbach, J. B. (2016), Journal of Labor Economics. Oster, E. (2019), Journal of Business & Economic Statistics.

PREDICT What happens to the standard error?

Adding Z changes the coefficient. Does its standard error go up or down?

ANSWER What happens to the standard error?

Both are possible. Two forces work against each other:

- **Down:** if Z explains part of Y, the residual variance falls and the estimate becomes more precise.
- **Up:** if Z is strongly correlated with X, less independent variation in X is left to identify β (collinearity).

$$\widehat{\mathrm{Var}}(\hat{\beta}^l) = \frac{\hat{\sigma}^2_{e'}}{\sum_i (X_i - \bar{X})^2 \, (1 - R^2_{X|Z})}$$

- The numerator falls with a good predictor of Y; the term $(1 - R^2_{X|Z})$ falls with collinearity.
- A control that predicts Y well and is unrelated to X (as in a randomised experiment) only helps.

STATA The identity on real data

```
sysuse nlsw88, clear                  // US women, 1988 (ships with Stata)
generate lnwage = ln(wage)

quietly regress lnwage collgrad               // short
scalar b_short = _b[collgrad]
quietly regress lnwage collgrad ttl_exp       // long: add total experience
scalar b_long  = _b[collgrad]
scalar gamma   = _b[ttl_exp]
quietly regress ttl_exp collgrad              // auxiliary
scalar delta   = _b[collgrad]

display "short = " b_short "  long = " b_long
display "short - long = " b_short - b_long "   gamma*delta = " gamma*delta
```

- Before running: do college graduates in this sample have more or less experience? So which way should the coefficient move?
- Then repeat with `tenure` instead of `ttl_exp`. Careful: `tenure` has missing values. Make all three regressions use the same observations (e.g. `if !missing(tenure)`), otherwise the identity fails.

Take-aways

- Adding a control moves the coefficient by exactly $-\hat{\gamma}\hat{\delta}$: down, up, across zero, or not at all.
- The direction is predictable from two signs: how Z relates to Y, and how Z moves with X.
- Before adding a control, ask *when* it is determined: before X (candidate good control) or after X (possible bad control).
- An unchanged coefficient is not evidence of no confounding; judge the movement against how much the controls explain.

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